



JINKA UNIVERSITY
COLLAGE OF NATURAL AND COMPUTATIONAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE
GRADUATION CREDENTIAL VERIFICATION SYSTEM FOR JINKA
UNIVERSITY (GCVSJKU)

By

- | | |
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| 3. Kedir Seid Ali | UGR/14168/14 |

Advisor: Mr Endalkachew Emire (MSc.)

Mr Senbetu Abebe (MSc.)

Jinka, Ethiopia
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A Final Year Project I Submitted to Jinka University, College of Natural and Computational Science, Department of Computer Science, in Partial Fulfilment of the Requirements for the Accomplishment of BSc. Degree in Computer Science.

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This project entitled “Graduation Credentials Verification System for Jinka University (GCVSJKU)” has been read and approved as meeting the requirements of computer science in partial fulfilment for the award of the degree of Bachelor of Science in Computer science, Jinka University, Jinka, Ethiopia.

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ABSTRACT

*Forgery has long crept into our educational system and counterfeit certificate(s) and document(s) have upper hand in various certificate and degrees awarding institution. Invalid studentship has become greater than valid studentship. The need for student verification system has become necessary to check mate and expunge invalid studentship, invalid document and invalid certificate yielding for the authenticity of certificate and documents. This project titled “**Graduates Credentials Verification System for Jinka University**”. The proposed system is concerned with verifying certificates (graduate, postgraduate) issued from JKU and processing several academic requests from the graduated students in the future. **Graduates Credentials Verification System for Jinka University** provides a comfortable communication among JKU and outside organization(s) as nation’s source for accurate degree data. Moreover, it will be a web-based system where the users will be provided with comfortable interfaces to interact with JKU for their immediate requirements in a trustworthy manner from anywhere. The system will have a great significance on the JKU as well as Ministry of Education in the future. This is an attempt from Ethiopians for the Ethiopians to obliterate the counterfeit from the educational systems.*

ABBREVIATIONS

CD	Compact Disc
EC	Ethiopian Calendar
GB	Giga Bytes
GCVS	Graduates Credential Verification Systems
HTML	Hypertext Mark-up Language
ID	Identification Number
JKU	Jinka University
JKUGCVS	Jinka University Graduates Credential Verification Systems
MySQL	My Structured Query Language
PHP	Hypertext Pre-processor
RAM	Random Access Memory
UC	Use Case
UI	User Interface
UML	Unified Modelling Language

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CHAPTER ONE

1. INTRODUCTION

A verification system as in this design is an operation program used to corroborate that someone or commodity correctly and fairly belongs to a collection or an association. There are different verification systems, especially address verification systems and card verification systems to mention but many. Credentials verification system is used to know whether or not the instrument handed is legal and correctly issued for the pupil. In this environment, it's also used to corroborate that a pupil belongs to an institution and to confirm that he or she's duly registered in a particular department running any of the institutions program and checking whether the pupil Credentials is correct indeed if their name is registered and instrument is given for them. This Credentials verification system, is an electronic and motorized means of vindicating someone's claim of studentship or validating degree instrument by an institution using an affair easily showing the verity or false of his or her claim.

1.1 Background of the Organization

Jinka University (IKU) is a Public Higher Education Institution (HEI) established in 2017 with regulation No. 353/2015 of the Council of Ministers of the Federal Democratic Republic of Ethiopia. JKU is one of the youngest fourth generation comprehensive universities in the country. Geographically, JKU is located in South-western Ethiopia of the Ari Zone in the vibrant town of Jinka at a distance of around 750 Kms from the capital city of Ethiopia, Addis Ababa. Currently, JKU runs 29 undergraduate programs under five colleges and one institute (1).

Over the past years, students' identification and verification has been a major problem in large institutions as documents, instruments and students can be forged at a great rate and easy way, using the computer negatively. Forgers fail to know that in this presto moving world, the computer has inversely served as an aid to decision timber, verification and authentication. This is because of computers effectiveness in terms of speed, delicacy, trust ability, cost and security among others. In recent times, mindfulness has been created in people as it concerns the use of computers in vindicating, authenticating and security conditioning either through web- grounded (internet) or window- grounded. The value of every degree is the character of the institution and

the scholars produced, hence, the topmost handicap to any academic institution which is deceitfulness and phony has to be checked with verification and authentication systems and processes. Certificate verification is mandatory for the most of the reputed companies or associations in order to reduce the pitfalls related to it. At most of the universities as well as educational providers come up with motorized verification systems and also support electronic verification. But Jinka University still now has no similar verification system that can help essential for the graduated scholars when they're hiring or hired in any organization. Also, enrolment verification is designed to fulfil the verification requirements of companies that offer products or services requiring proof of a student's enrolment status. For accessing several academic documents graduated students spends much time and cost and loose several good career opportunities in their life.

1.2 Related work

Prior to beginning this study, there were numerous studies conducted at various universities, including the 2016 GC work by Gebremedhin Mekonnen and Samson Endale at Bahir Dar University's Faculty of Computing, Computer Science, and Engineering (Software Stream). This system exclusively targets higher education institutions in Ethiopia (2). The gap is the system cannot view the full certificate and is unable to print it.

1.3 Statement of the Problem

At present Jinka University isn't having any computerized system to deal with degree verification for maids as well as post graduate degree holders from then. Only thing is the employer or whoever concerned requirements to communicate the office of register or academic record office for similar purposes can communicate the register and also get the pupil Credentials. This is one of the great impacts on Jinka University and also the country.

- Cost and time consuming.
- May be falsifications of results by the office
- Vulnerable to different factors such as fraud certificate and corruption.
- Lack of security in existing system.
- letter lost during the postal service's sending and receiving process

- Tedious and less speed in the process of verification.
- Illegal or faulty modification of credentials.
- Fault certificate made by Jinka University name.
- Increased illegal employee in the country.
- Declined legal credentials values
- Embarrassment and negative impact to the organization's reputation.

1.4 Objectives

1.4.1 General objects

To develop a graduation credential verification system for Jinka university.

1.4.2 Specific objects

To achieve the over mentioned general idea, the design addressed the following specific objects.

The system provides the following

- To prepare user friendly website where specific action for registrar, administrator and external users such as organizations are defined.
- To store and manage graduated student's information.
- To give proper administration features like: user management and authentication.
- To provide options to download or email verified student information including degree and post graduate certificate in pdf format.
- Online application for various service required by the organization.
- To design and implement a database for proper implementation of the system.
- To search graduated student's file.
- To allow students to print a duplicate certificate using their certificate ID if it is lost.
- To implement a feature for remote printing of certificates, especially for students who are away from the university.

1.5 Scope and Limitations of the Project

1.5.1 Scope of the Project

The scope of this project focuses on the online Graduated credentials verification system for JKU and authentication of documents only for bachelor's degree and postgraduate degree. Thus, graduates must be graduated from JKU and also the system is available only for JKU.

1.5.2 Limitation of Project

- The system will be developed for Jnka University.
- The system is only for graduated students.
- The system only supports English languages.
- The system depends only on network.
- Website promotions about other websites will not be performed. It does not include any promotion of website.

1.6 Methodology and Tools

1.6.1 Data Collection Methodology

Data collection is one of the important tasks in determining and analysing the existing system. The data that we need to develop the new system are obtained from system users, forms and documents in the Jinka University main registrar.

To gather data from the different users and manager of the existing system project team used the following techniques: -

- **Interview:** Different users of the existing system along with managers were involved to have an understanding about how the system works, what are its different features, factors, and any disadvantages as well as advantages so we obtain detailed information from them.
- **Direct Observation:** We observed various online verification systems and academic request service providers on the internet (2) . We also observed the existing academic

credentials request system to know how the process is going and to understand the management features it provides.

- **Document Analysis:** To get information about the background of the system project team review necessary documents, books that support this project. To make this project more feasible and structured the team are analysing other thesis management websites from universities other in our country and others, and the internet for explanation about some word definitions and for search books.

1.6.2 System Analysis and Design Methodology

Among the different methodologies available we're using object acquainted methodology for the analysis and design of our system. Object acquainted methodology enables us to represent complex relations among different objects and represent data and process with harmonious memorandum throughout the system. An important point similar as heritage, polymorphism, data caching and encapsulation suits the design veritably well. This greatly improves the communication among guests, analysers, contrivers and programmers. It increases thickness among analyser, developer, perpetration and testing. It also allows the re-usability of the law which will help to enhance the design in the future.

To develop this system, the waterfall model system development methodology is usable. Because of several reasons, choose waterfall; some of them are:

- **Structured Approach:** The Waterfall model provides a clear, linear progression through distinct phases (requirements, design, implementation, verification, maintenance), facilitating easier project management and tracking.
- **Documentation:** It emphasizes comprehensive documentation at each stage, ensuring that all requirements and design decisions are well-documented, which aids in future maintenance and knowledge transfer.
- **Predictability:** The sequential nature of the Waterfall model allows for better predictability in timelines and deliverables, making it easier to estimate project duration and resource allocation.

- **Simplicity:** The model is straightforward and easy to understand, making it suitable for projects with well-defined requirements and minimal expected changes.

In contrast, OOP focuses on the design and organization of code through objects and classes, which may introduce complexity and require more iterative processes.

Water fall model was the first process model to be introduced. It also pertains to a direct-successional life cycle model. In a cascade model, each phase must be completed completely before the coming phase can begin. At the end of each phase, a review takes place to determine if the design is on the right path and whether or not to continue or discard the design. In this model the testing starts only after the development is complete. In cascade model phases, they don't overlap. The waterfall approach is preferred for the following advantages:

- This model is simple and easy to understand and use.
- It's easy to manage due to the severity of the model—each phase has specific deliverables and a review process.
- In this model, phases are reused and completed one at a time. Phases don't lap.
- It has a clear structure and phases.
- The waterfall model follows a linear and sequential approach, with distinct phases.

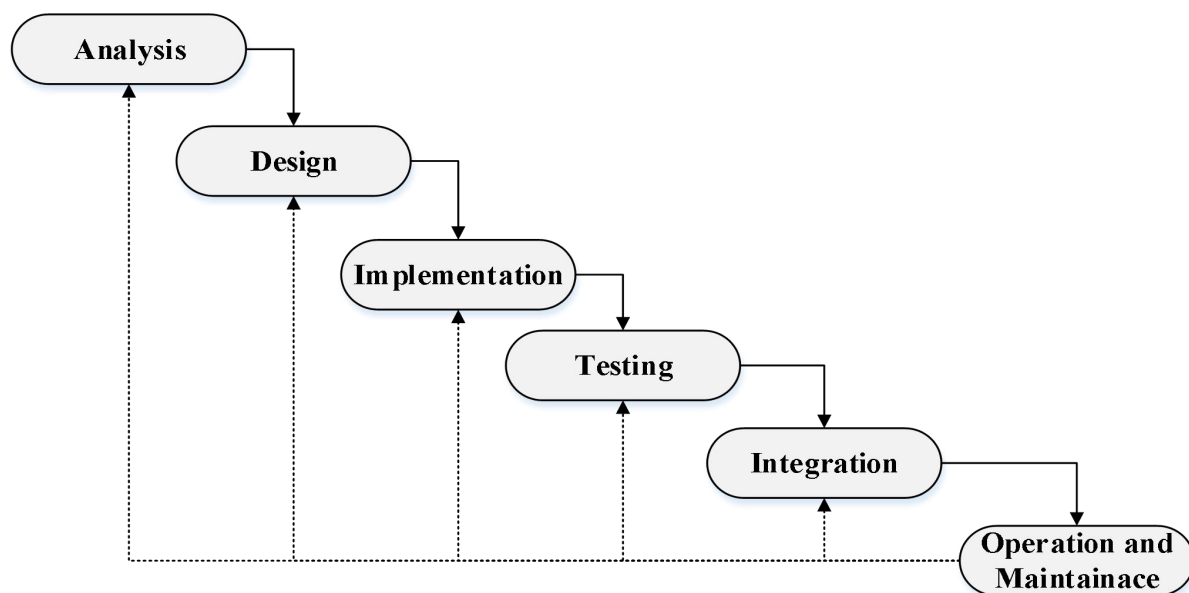


Figure 1.1 Waterfall model of system design

1.6.3 System Development Tools

We use the following software and software tools to develop our system. Some of these listed as follows:

1.6.3.1 Hardware Requirement

Activities	Tools
Training	Papers
Writing Something	Pen
Writing Soft copy of Documentation Part	Computer (Desktop or Laptop)
Printing Purpose	Printer
Storing File	CD
File Transferring	Flash 64 GB
Writing Something	Pencil
Binding Purpose	Binder

Table 1.1 Hardware Requirement

1.6.3.2 Software Requirement

Activities	Programs	Why
Code Writing	Dream-weaver, Notepad++	Easy to use
Database Server	XAMPP Server	Database server for PHP
Web Server	Apache (IIS)	Web server for PHP
Server-side Scripting	Visual Studio Code	Understand by server
Client-Side Coding	Visual Studio Code	It works with PHP
Client-Side Scripting	Visual Studio Code	Easy way to validate form
Coding Language	Visual Studio Code	Easy to writing code
Cascading Style Sheet	Visual Studio Code	It works with HTML and PHP
Browsers	IE, Mozilla Firefox, Google Chrome, Spark browser	Better for displaying web pages
Documentation	MS Word 2010	Easy to use
User Training	MS PowerPoint	Best for presentation
Modelling and Scheduling	Edraw, Visio 2013	Easy to design diagrams

Table 1.2 Software Requirement

1.7 Significance of the Project

The benefits of the design are listed below

- Cost and time reduction.
- Minimize defective instruments.
- Avoid revision made by scholars on their credentials.
- Increased the speed of access or faster credential verification.
- Reduction in material operation.
- Increased delicacy and vacuity.
- Increased quality of education throughout the country.
- Minimize corruption.

- Greater convenience.
- Allows penetrating the database fluently.
- Advanced fraud discovery.
- To make timely and well- informed hiring, creation and compensation opinions in an association.
- Avoiding data loss because of an indecorous data storehouse.
- Reduce force.

1.8 Project Management Technique

1.8.1 Project Budget

Resources like time and equipment are needed for any system development. Some of resources needed for project team are listed below with their estimated cost.

No.	Item Description	Quantity	Unit	Unit Price	Total price
1	Removable flash disk	2	Birr	600.00	12000.00
2	Binding	5	Birr	20.00	100.00
3	Paper	2 pack	Birr	1000.00	200.00
4	Pencil	5	Birr	10.00	50.00
5	Pen	10	Birr	25.00	250.00
6	Printing	1	Birr	165.00	165.00
7	hp Laptop	2	Birr	80000.00	160000.00
8	Transportation		Birr	5000	5000
9	Wi-Fi and Data		Birr	15000.00	15000
Total					181965.00
Contingency					10000.00
Grand Total					191965.00

Table 1.3 Project Budget

1.8.2 Project Schedule

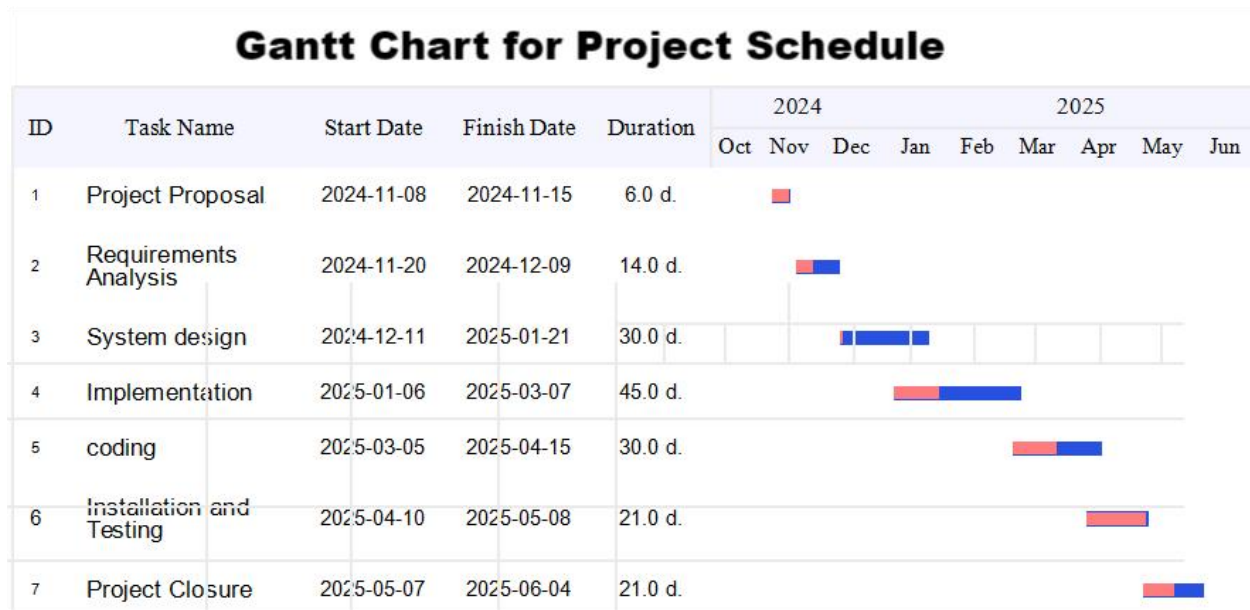


Table 1.4 Gantt chart for Project Schedule

1.9 Feasibility study

Feasibility analysis is an analysis and evaluation of a proposed project to determine if it is technically feasible, feasible within the estimated cost, feasible within political and feasible within operation (5)

1.9.1 Economic Feasibility

The purpose of assessing economic feasibility is to identify the financial benefit and cost association with the development project, economic feasibility is referred to us cost benefit analysis.

There are two types of benefits:

1. **Tangible Benefits:** The system can avoid unnecessary costs, for example, for paper, pen, transport, and payment for the post office if the process is through post offices.

2. **Intangible Benefits:** The cost associated with an association with an information system that cannot be easily measured in terms of birr or with certainty. Intangible benefits are as follow: -

- Improved user satisfaction
- Time saving
- Decision making based on quality information etc.

1.9.2 Technical Feasibility

The system can be technically feasible because the technical resources needed to develop, install and to operate are available in the present infrastructure. It doesn't require any technical expertise and can be integrated with the existing system.

1.9.3 Political Feasibility

The system provides online verification and manages services related to graduated students, which is more convenient to use for graduated students and registrar workers in JKU. This will provide good communication among the outside organization and JKU. The system "JKU GCVS" will be a significant contribution to the development endeavours of our country and supports the rule and regulation of the government hence it's politically feasible.

1.9.4 Operational Feasibility

The system can be used effectively after it has been developed. Users will not have any difficulty with the new system, producing the expected benefits. System will be developed based on the policies of the university plus the new business rules.

CHAPTER TWO

2. REQUIREMENT ELICITATION

2.1 The over view of the existing system

Existing system is using manual way of performing any activities for many processes such as:

- Verifying graduate by phone communication
- Verifying graduate by sending official transcript
- Verifying graduate through stamp and signature

Verifying graduate by phone communication: - the graduates are going to the organization to hiring and the organization manager will verify the graduate by calling phone who graduates of the institute.

Verifying graduate by sending official transcript: -the graduate wants to hire to the organization and the organization requests the graduate official transcript. Then the graduate must be sending the official transcript from their institute.

Verifying graduate through stamp and signature: - the institutes are verifying the graduate certificate through their institute stamps and signature.

2.1.1 Problem of the existing system

The existing has different problem to process the information of guidance and about historical places. These are:

- The reporting system is not accurate.
- Registration of the customer is manual way.
- The absence of getting full information

2.1.2 Weakness of the existing system

- letter lost during the postal service's sending and receiving process
- Tedious and less speed in the process of verification.
- Cost and time consuming.
- Fault certificate made by Jinka University name.
- Embarrassment and negative impact to the organization's reputation

2.1.3 Strength of the existing system

- Verification qualifications
- Make a strong relationship between the organization and the company.
- Give the organization a good image to people and external companies.
- Create strong transparency between those of the organization and the company.
- Create a good accountability culture for the workers on the system.
- Develop organization reputation.

2.1.4 Business rules of the system

The system's business rule requires that the graduated student complete the following requirements:

Rule 1: Graduates must attend Jinka University

Rule 2: Graduates have a GPA of at least 2.0.

Rule 3: He or she must receive a score of at least 50 out of 100 in order to pass the exit test.

Rule 4: The graduates are required to pay their cost sharing, which is voluntary.

2.2 Overview of the proposed system

The GCVS JKU project aims to streamline the verification process for Jinka University graduates, enhancing the security and reliability of academic credentials. Through careful planning, data collection, and user-centred design, the project seeks to address the needs of graduates and organizations while ensuring a robust and efficient verification system.

2.3 Functional Requirement

Functional requirements drive the application architecture of a system. A requirement specifies a function that a system or component must be able to perform. Functional requirements specify

specific behaviour or functions (2) (3) . The functional conditions of the proposed system are listed as follows: -

- Authenticate and authorize users according to user's role.
- The system should check all the supplied user input.
- Enables directors to manage the druggies.
- The companies or associations must register their own biographies to corroborate instruments.
- The system should give the installation to view detailed information, add, edit, and cancel graduates' information for the druggies similar as director or director of register.
- Provide operation form, same as the being system, for the graduates to apply for academic credentials.
- Provides installations to download the academic credentials as per the requests from the graduates.
- The companies or associations can enter requests but cannot authorize or cancel requests.
- Members of the Registrar group can authorize a request but cannot cancel requests.
- Members of the director's group cannot authorize requests but can cancel requests.
- The admin can fit, modernize and search graduates' information.
- Data redundancy is avoided, through the proper design of the database.
- The system must give help to the user.

2.4 Non-Functional Requirements

Non-functional requirements describe user-visible aspects of the system that are not directly related with the functional behaviour of the system that are listed above. A non-functional requirement is a statement of how a system must behave; it is a constraint upon the system's behaviour. They specify criteria that judge the operation of a system, rather than specific behaviour (4). Non-functional conditions of the proposed systems are listed as follows: -

- **Availability:** The system available to the druggies 24 hours if internet connection and electric power is live.
- **Performance:** The system shall perform its operations within a minimal quantum of time. The system shall be interactive and the user gets the anticipated result within many

seconds. The system will be penetrated by numerous druggies inside or outside the university. For better response and outturn it should be given further emphasis for the briskly access to the system across the network. Constraints can be a user's computer low RAM, Network or connectivity failure, and others.

- **Security and Access permissions:** Administration part of the system is accessible only for the authorized Administration staff according to their Assigned right. An Unauthorized user shouldn't pierce the administration runner by any means. Some of the coffers and exertion shall be confined and allowed only for authorized druggies. Thus, the system shall have a user identification system and it should grant or deny access to druggiess as per the honour of the prospective user. There may be some interferers who could pierce the train or data on the database, but with the proposed system being part of JKU garcon this shall be a veritably rare occasion.
- **Deployment:** Final system shall be stationed in a web garcon (Apache) part of JKU.
- **Documentation:** The conditioning and labours of each system development stage in the design work inflow will be duly proved for the successful completion of the design. These documents produced at the end of each stage shall be organized and collected together at the end of the design for future reference, system conservation, and system support.
- **Extensibility:** Adding features and carry- forward of customizations at coming major interpretation upgrades along with the businesses-engineering is possible.
- **Provisory:** There shall be a frequent and full back over medium to avoid any information loss and inconsistency.
- **User interface:** The user interface of the system shall be user friendly and easy enough to work with. System provides web interfaces that are compatible with cyber surfers like Internet discoverer, Mozilla Firefox, Google chrome.
- **Communication Interface:** JKU GCVS will be connected to the original area network of JKU.

2.5 Constraints and Assumptions

2.5.1 Constraints

Constraints are rules or protocols that guide the system and users. Constraints can be classified into two types:

1. **Business constraints:** a constraint that applies to the operations of functions of the system.
 - Lack of money as we students.
 - Lack of our knowledge.
 - Lack of our skills.
 - Lack of time according to our corresponding class and course project.
2. **Technical constraints:** a constraint that applies during practical cases.
 - Due to connection problems
 - Failure of electricity without saving the files.

2.5.2 Assumptions

Assumptions are ideas, predictions about the system become true.

- Administrators control all admins in the system.
- Administrators and registrars get a notification sign from an external user; they request it.
- External users can easily access the system without the high experience required.

CHAPTER THREE

3. SYSTEM MODELING

This chapter covers the functional aspect of analysis that is functional modelling. The set of functional requirements that have to be analysed is done in an object-oriented way. UML (Unified Modelling Language) an object-oriented, graphic language, is unified for all the aspects of the requirement analysis. The functional model consists of use case modelling to show the system functionality from user point of view, use case description, activity diagram to describe the details of internal system functioning and sequence diagram, collaboration diagram and state chart diagram.

Actors	Use case
➤ External user ➤ Administrator ➤ Registrar	➤ Login ➤ Request Certificate Verification ➤ View Certificate Verification Link ➤ Manage Graduate Information ➤ Create and Manage Account ➤ Registration of Employee ➤ Approve Service Request

3.1. Actor and Use Case Identifications

Table 3 1 Actor and use case identification

3.2. Use case Diagram

Use case diagrams are ideal artifacts to capture the requirements of the system. They represent the user requirements gathered during requirement elicitation. Use case diagram consists of use cases, actors, and their relationship and system boundary (6).

When we draw a use case we use the following shapes.

No	Shapes	Representation
1		It represents the users of systems known as actors . It represents a role that a user can play but not a specific user.
2		It represents a user goal that can be achieved by accessing the system or software application.
3		It represents the scope of systems. Known as the system boundary. The use case of a system is placed inside this, while the actors who interact with the system are put outside the system.
4	--<<include>>-->	It represents that a use case contains the behaviour defined in another use case.
5	--<<extend>>-->	It represents how and when the behaviour defined in the extending use case can be inserted in to the behaviour defined in the extended use case.
6		It is used to associate an actor and use case to indicate that the actor participates in that use case. Therefore, an association corresponds to a sequence of actions between the actors and use case in achieving the use case.

Table 3.2 Use case representations

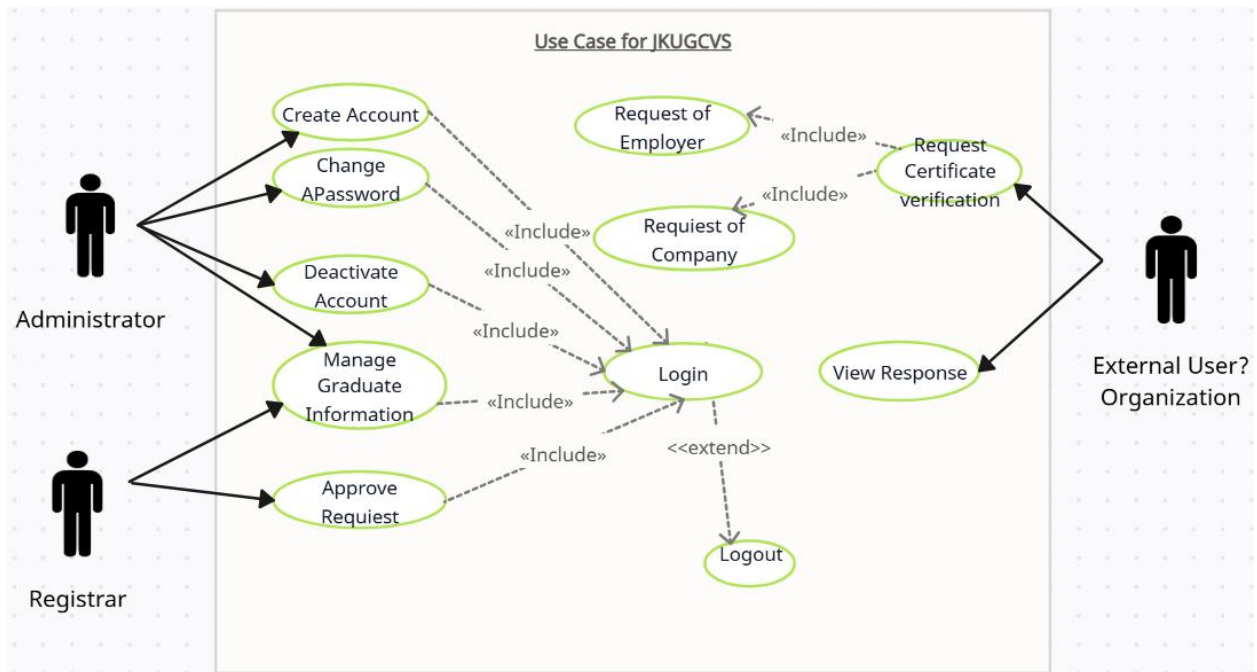


Figure 3.1 Use case diagram

3.3. Use Case Description

Use Case ID	Use Case Name	Description
UC_01	Log In	It describes the process of authenticating the user Name & password before granting access to his or her GCVS account.
UC_02	Request Verify Certificate	It describes the process of starting verifying the graduated student's credentials.
UC_03	Manage Graduate information	It describes the process of updating, inserting, removing, searching data or files related to the graduate's credentials.
UC_04	Approve Service Request	It describes the process of verifying & approving the service requests of the organizations or companies by the director of registrar.
UC_05	Create or Manage Accounts	It describes the process of creating or managing account for the authorized users of the system like director of registrar.
UC_06	Change Password	This use case allows the Administrator to change the password.
UC_07	Deactivate Account	This use case allows the Administrator to delete account.
UC_08	Registration of Employer	It describes the process of registration of the employer and company or organization details with the university in order to verify the candidate's credentials.
UC_09	View Certificate Link	It describes the process of viewing a certificate verification that sends to external user throw their email.

Table 3.3 Use case descriptions

Use Case ID	UC_01	
Use Case Name	Log in	
Primarily Actor	Administrator, Registrar.	
Description	It describes the process of authenticating the user's name & password before granting access to his or her GCVS account.	
Trigger	Administrator, Registrar and need to login his or her account to use this system.	
Pre-Condition	Administrator and Registrar login.	
Normal Action	Actor's Action ✓ Step 1: Administrator and Registrar initiate to login. ✓ Step3: Administrator and Registrar enter a valid user name & password.	System Response ✓ Step 2: The system displays the login page. ✓ Step 4: System checks the authentication of username and password. ✓ Step 5: System displays their corresponding GCVS user page.
Post condition	Administrator and Registrar can do different activities depending on the permission provided to them.	
Alternative Action	✓ Step 4: If the username and password is not valid, the system displays an error message and goes to step 2.	

Table 3.4 Use case descriptions for login

Use Case ID	UC_02	
Use Case Name	Request Verify Certificates.	
Primarily Actor	External User.	
Description	It describes the process of requesting to verify the graduate's credentials.	
Trigger	External Users want to request to verify their credentials.	
Pre-Condition	Companies should register themselves and graduate profiles.	
Normal Action	Actor's Action ✓ Step 1: External User initiates to requesting verification. ✓ Step 3: External User enters the organization details and graduates' profiles.	System Response ✓ Step 2: The system displays the Requesting form. ✓ Step 4: Systems validate information submitted. ✓ Step 5: System sends the certificate verification link through the email. ✓ Step 6: System stores the user's information and activity.
Post condition	External user read Certificate Verification link	
Alternative Action	✓ Step 4: If entered information is not valid the system displays an error message and goes to step 2.	

Table 3.5 Use case description for verify certificates

Use case ID	UC_03	
Use case name	Manage Graduate student information.	
Actor	Administrator, Registrar.	
Description	It describes the process of updating, Searching, adding, removing data or files related to the graduated student's files or credentials.	
Trigger	Administrator or Registrar wants to Searching, updating, delete, add information.	
Precondition	Administrator or Registrar has to be remembering username and password.	
Normal Action	Actor's Action (for update) ✓ Step 1: Admin or Registrar initiates Searching information. ✓ Step 3: User clicks Manages Graduates information link. The user can update, search, delete, view, and insert their task. ✓ Step 4: Here the user clicks the search option. ✓ Step 6: User enters the student ID and clicks OK. Step 8: User View information Searched.	System Response ✓ Step 2: System displays Manages Graduates information link. ✓ Step 5: System asks to enter the student ID. ✓ Step 7: System displays the searched information about the student.
Post condition	User uses the searched information.	
Alternative Action	✓ Step 9: If the ID entered violates the data entry rules, searching fails and informs the user.	

Table 3.6 Use case descriptions for manage graduates' information

Use Case ID	UC_04	
Use case name	Approves Verification Service Request	
Actor	Director of Registrar	
Description	It describes the process of verifying & approving the Verification service requests of the organization or companies by the director of registrar.	
Trigger	User wants to check and approve Verification service requests of Organization or companies.	
Precondition	User has to login.	
Normal Action	Actor's Action ✓ Step 1: User initiates for approving requests. ✓ Step 3: User clicks the List for approval of service request link. ✓ Step 5: Users enter approved remark and clicks to generate report based on the request. ✓ Step 9: User views the report and enters his remark and clicks save.	System Response ✓ Step 2: System displays the List for approval of service request link. ✓ Step 4: System displays the list of Organization or Companies requests for different Verification services, along with the fee details submitted by students. ✓ Step 6: System updates the database, with the user's remark. ✓ Step 7: System generates requested document. ✓ Step 8: System displays the report for the user with option for further entry option. ✓ Step 10: Systems save the report to the database and sends credential verification link to organization.
Post condition	External user or organization views the certification Verification link.	
Alternative Action	✓ Step 5: If confirmation fails, user updates the database with rejected remark, followed by reason. Mail is sent to the requested External user or organization.	

Table 3.7 Use case description for approve verification service request

Use Case ID	UC_05	
Use Case Name	Create or manage Account	
Actor	Administrator	
Description	It describes the process of creating a new mail account for the users by the administrator.	
Trigger	Administrator needs to create an account for the user to use the system.	
Pre-Condition	The Administrator who knows the administrator password should login as and use the create account facility in the system.	
Normal Action	Actor's Action ✓ Step 1: The Administrator initiates to login. ✓ Step 3: Administrator enters valid user name & password. ✓ Step 6: From the admin page Create Account Option is selected by the Administrator. ✓ Step 8: System administrator enters user information.	System Response ✓ Step 2: The system displays the login page. ✓ Step 4: System checks the authentication of username and password. ✓ Step 5: System displays admin page. ✓ Step 7: System displays create account page. ✓ Step 9 System creates user account.
Post condition	New user account is created which the user can use for different activities.	
Alternative Action	✓ Step 4: If the username and password is not validated and verified, the system displays an error message and goes to step 2. ✓ Step 8: If the user information is not validated and verified, the system displays an error message and goes to step 7.	

Table 3.8 Use case descriptions for create or manage account

Use Case ID	UC_06	
Use Case Name:	Change password	
Actor	Administrator	
Description	This use case allows the Administrator to change the password.	
Trigger	Administrator needs to change password of the user.	
Pre-condition	The administrator login to the system and there should be account with password to be changed.	
Basic Course of Action	Actor action Step1: This use case starts when the administrator browses change password. Step3: Then he/she fills the necessary information to change password. Step6: Use case ends after the confirmation is sent for the administrator.	System response Step2: The system displays the change password form. Step4: The system validates the inserted information. Step5: The system notify as the password is changed.
Alternative Course of Action	A4: The system determines the invalidation of the inserted information and notifies to re-insert information again.	
Post condition	The password with the given account will be changed.	

Table 3.9 Use case descriptions for change password

Use Case ID	UC_06	
Use Case Name	Deactivate Account	
Actor	Administrator	
Trigger	Administrator needs to delete account of the user.	
Description	This use case allows the Administrator to delete account.	
Basic Course of Action	Actor action Step1: This use case starts when the administrator selects deactivate account button. Step3: Then the administrator fills the necessary information. Step6: Use case ends after the confirmation is sent for the administrator.	System response Step2: The system displays the deactivate account form. Step4: The system validates the inserted information. Step5: The systems notify the confirmation the account is deactivated.
Alternative Course of Action	A4: The system determines the invalidation of the inserted information and notifies to re-insert information again.	
Pre-condition	The administrator login to the system and there must be account to be deleted/deactivated	
Post condition	The account deleted/deactivated.	

Table 3.10 Use case descriptions for deactivate account

3.4. Sequence diagram

When we draw a sequence diagram we use the following shapes.

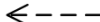
No	Shapes	Representation
1		It describes the way an object will behave in context
2		Known as activation box, used to represent the time an object needs to complete a task.
3		It is a message that represents communication between objects.
4		Lifelines that represent the object's presence overtime
5		Reply message that represents message is back to the original lifeline
6		A message that an object sends to itself.
7		Delete icon that represents destruction of object.

Table 3.11 Shapes of sequence diagram

Sequence Diagram for Login

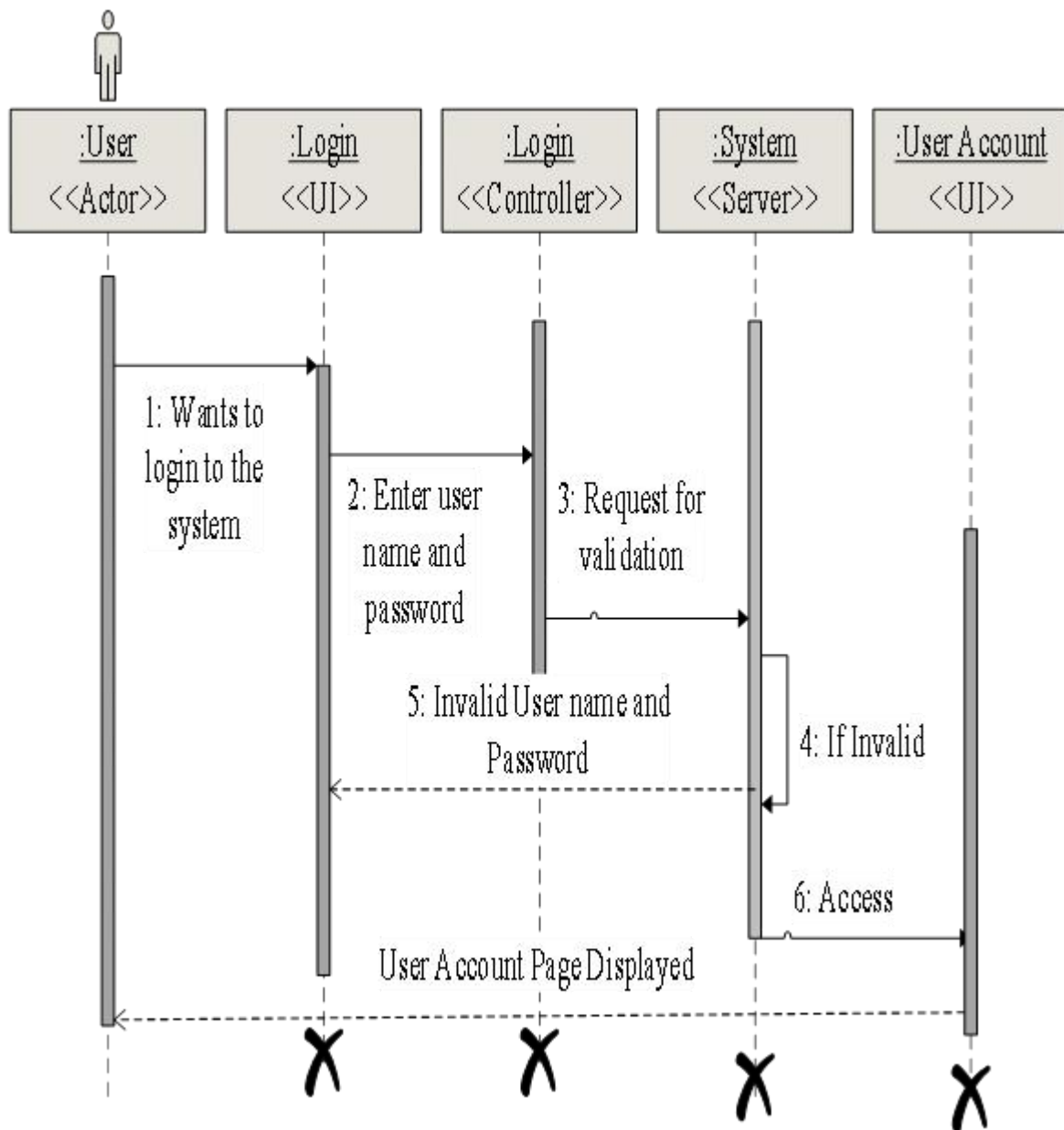


Figure 3.2 Sequence diagram for login

Sequence Diagram for Requesting Academic Service

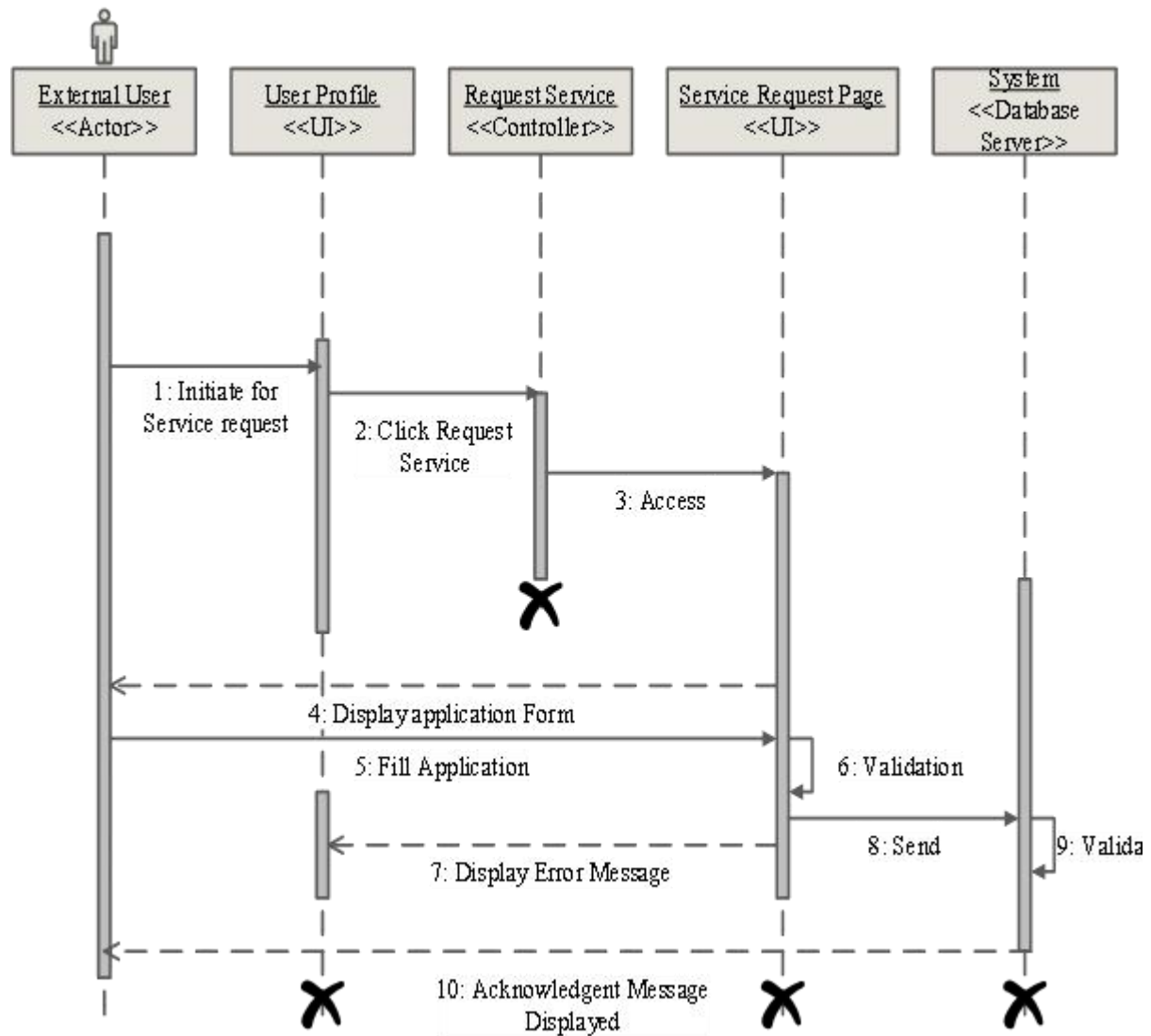


Figure 3.3 Sequence diagram for requesting academic service

Sequence Diagram for Approving Services

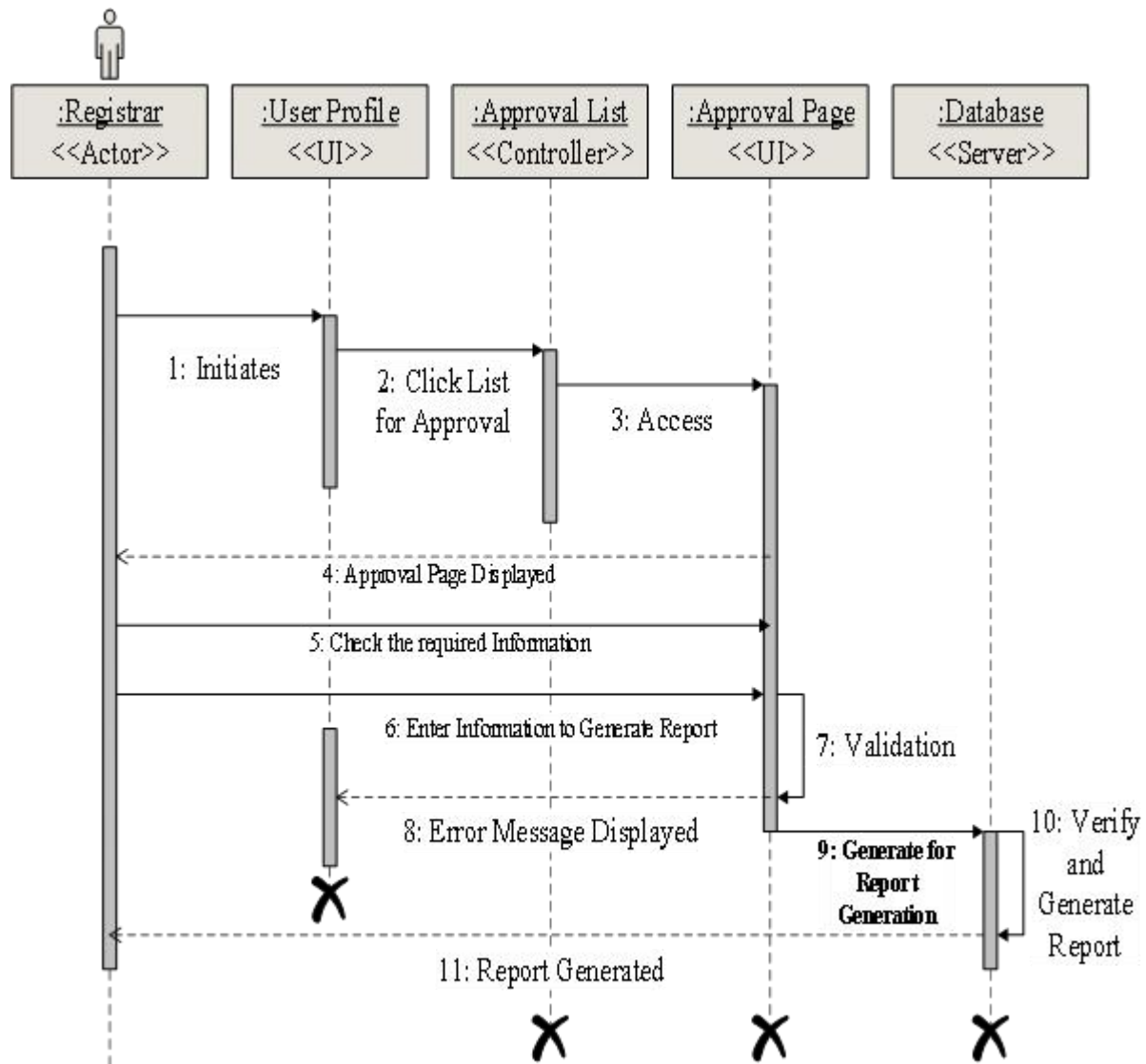


Figure 3.4 Sequence diagram for approving services (for credentials from graduates)

Sequence Diagram for Verify Certificates

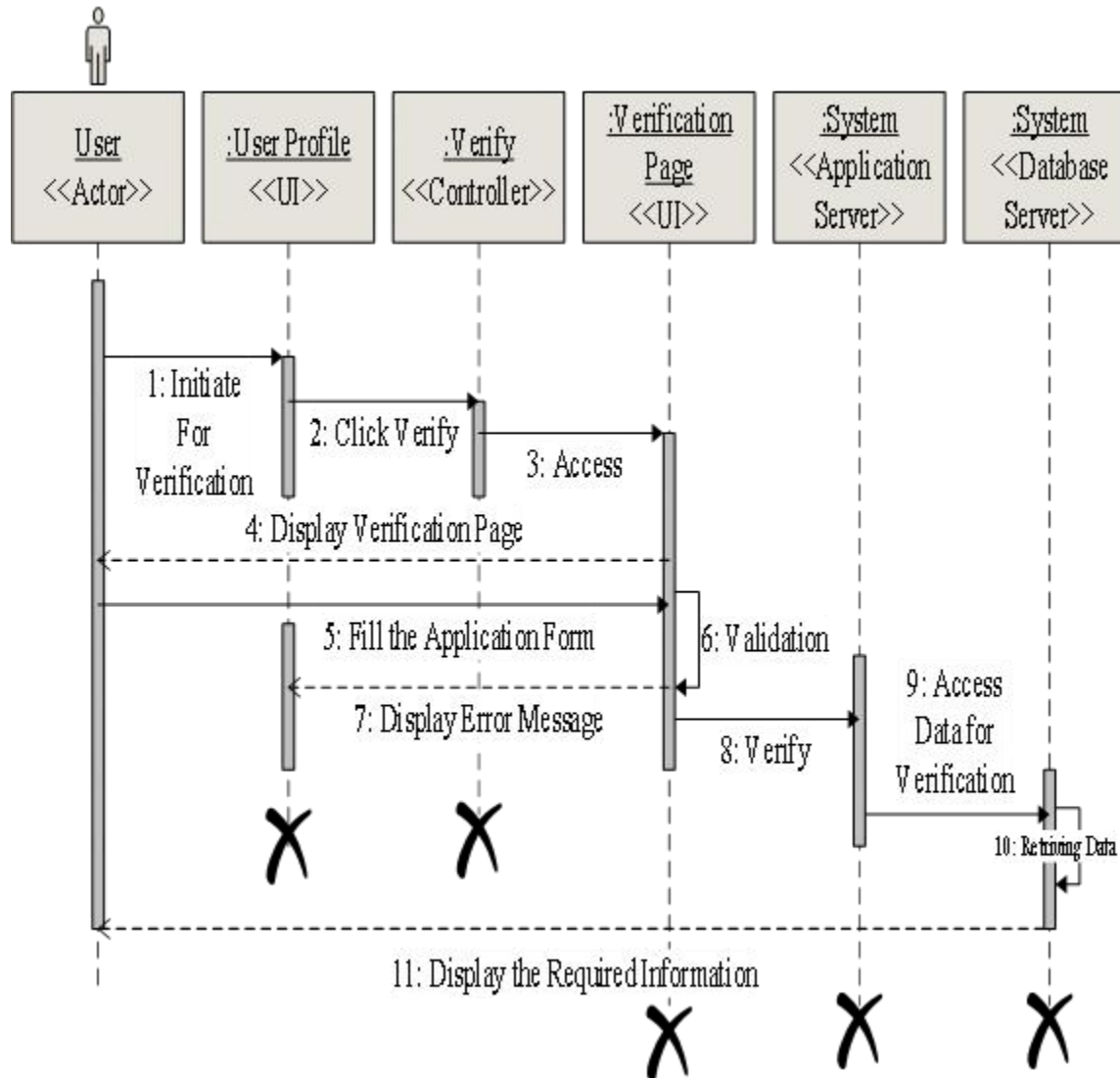


Figure 3.5 Sequence diagram for verify certificates

Sequence Diagram for Searching Graduates Information from the Database.

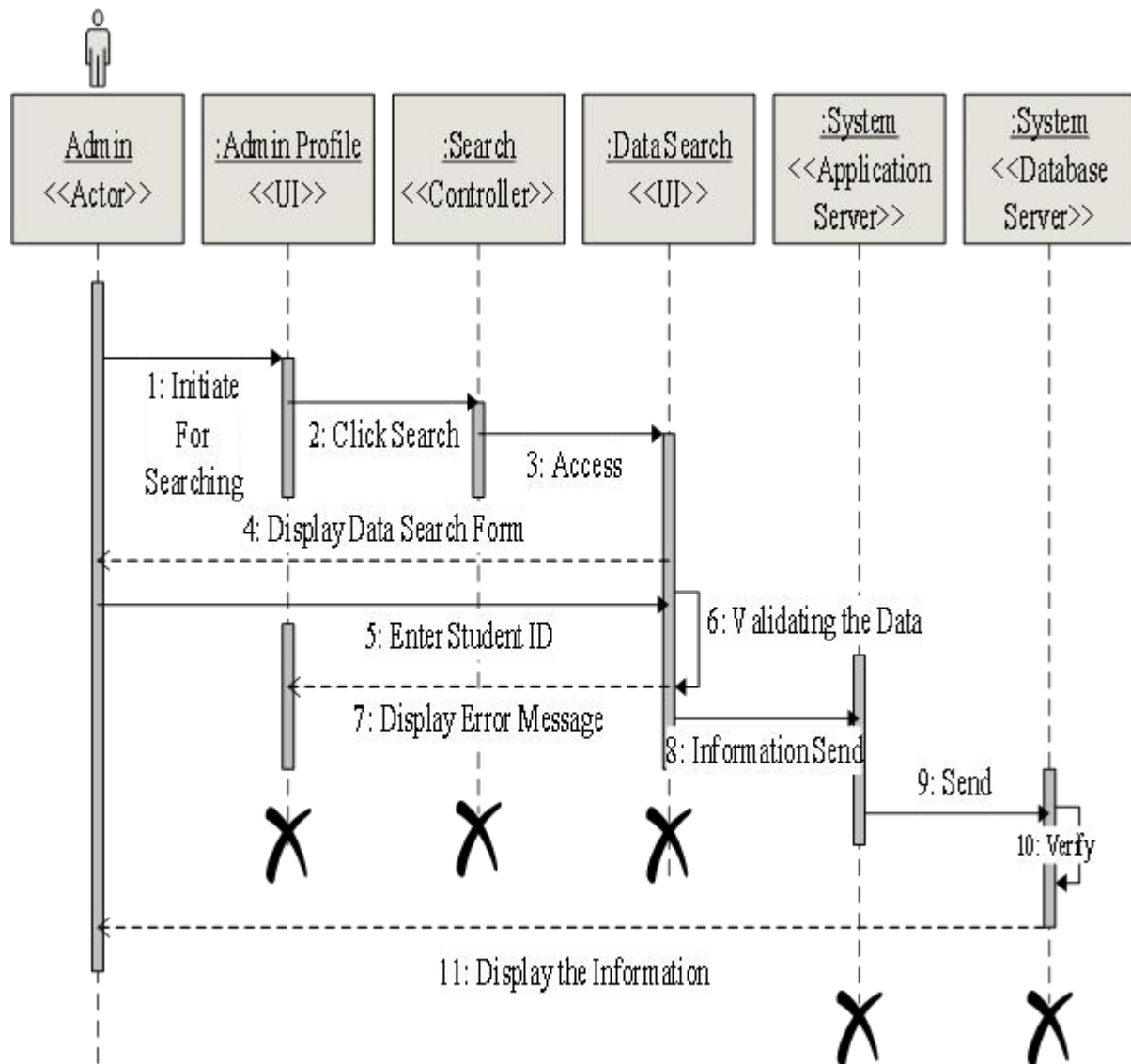


Figure 3.6 Sequence diagram for searching graduates' information from the database

Sequence Diagram for Create Account

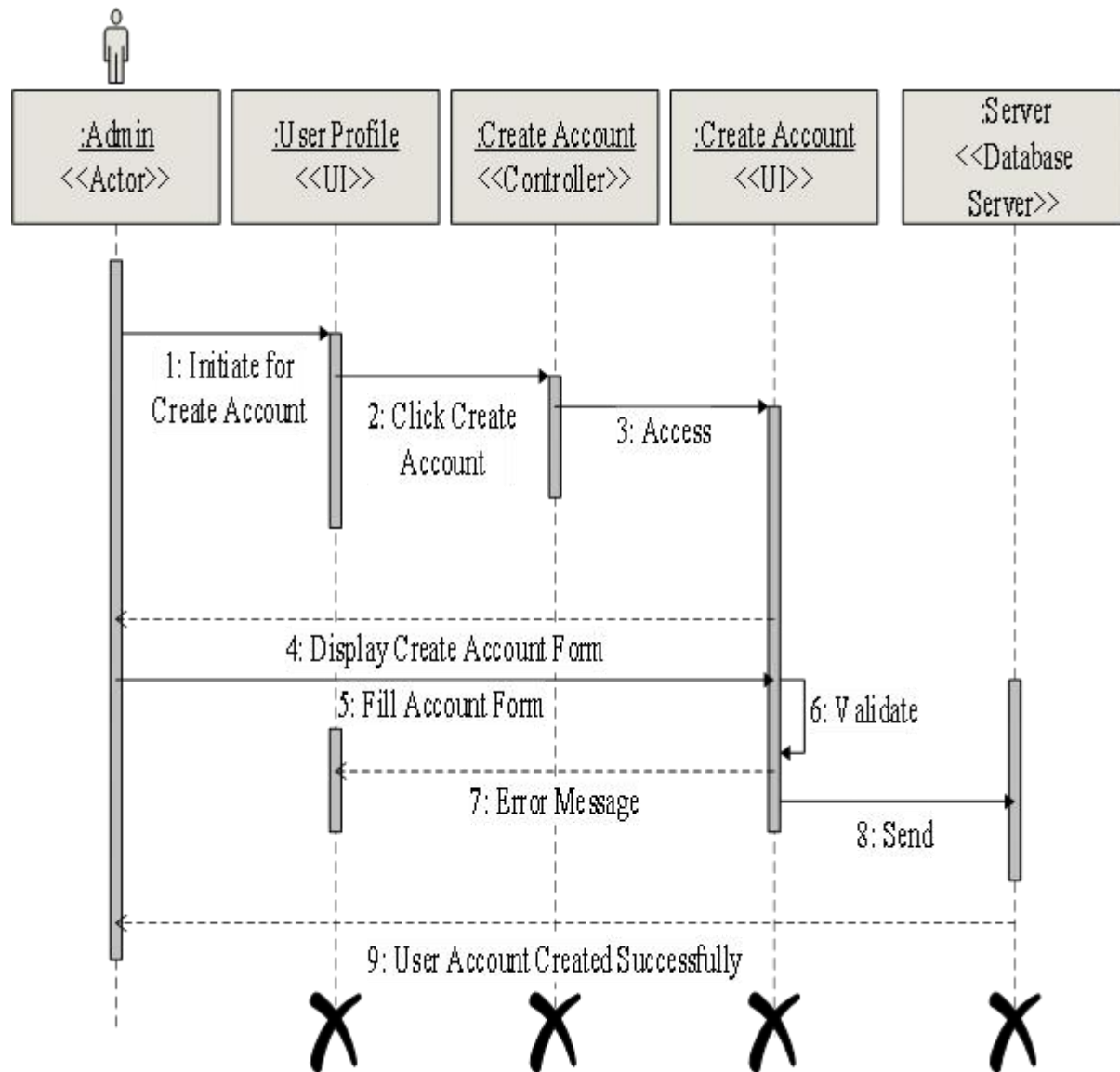


Figure 3.7 Sequence diagram for create account

3.5. Class Diagram

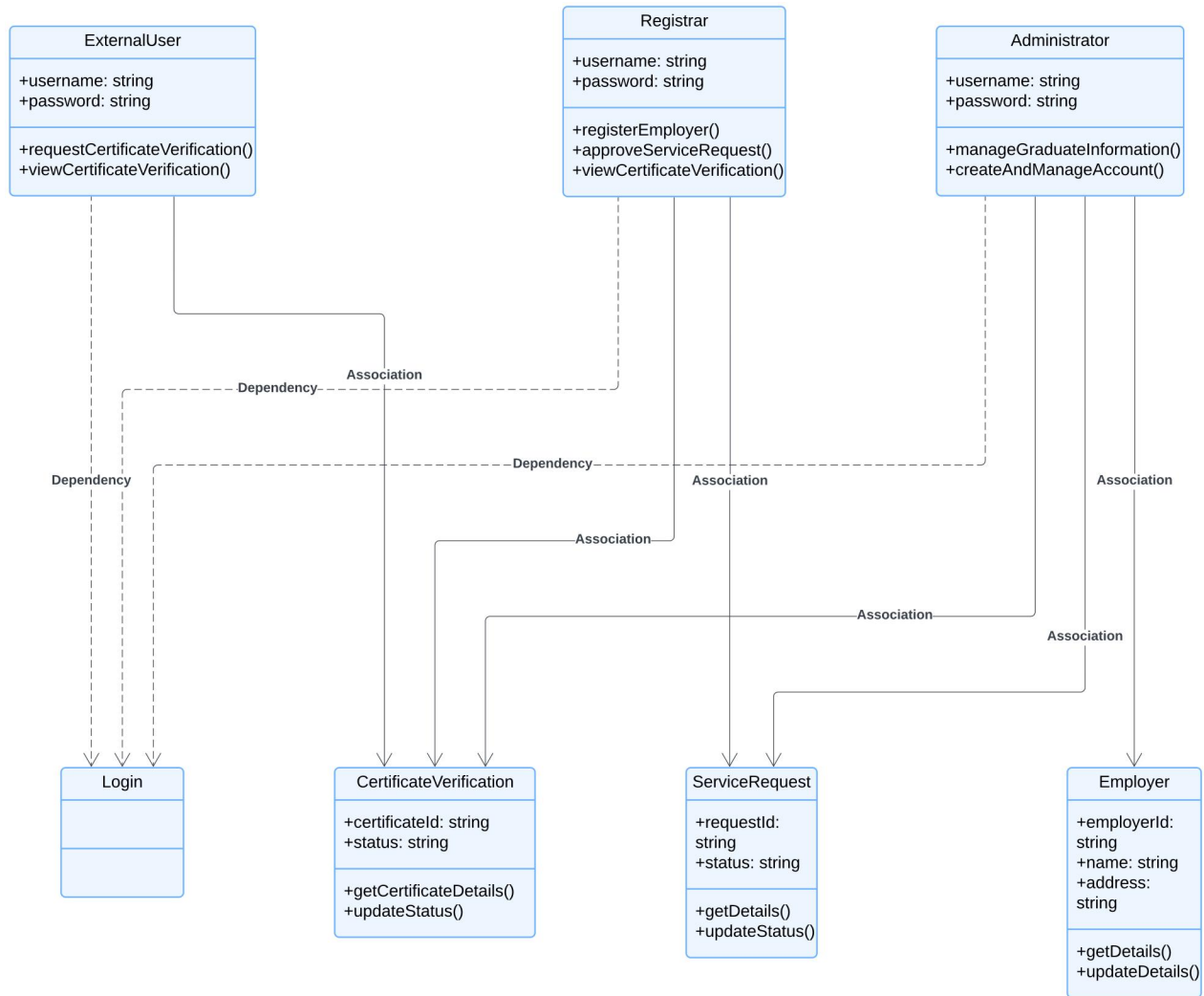


Figure 3.8 Class Diagram

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